# ARGOFLOW PRO X9

Support & Troubleshooting Procedures
Field Service and Diagnostics Reference

Veltrion Automation Systems — Technical Support Division
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# 1. How to Use This Guide

This document provides structured troubleshooting procedures for the ArgoFlow Pro X9 industrial IoT edge controller, intended for use by site technicians, systems integrators, and Veltrion-certified support partners. It should be used alongside the companion Product Guide (VAS-PG-X9-R4), which covers installation and configuration.

Each troubleshooting section follows a consistent structure: a description of the symptom, the likely root causes in order of probability, and a step-by-step resolution procedure. Where a step requires escalation beyond field-level troubleshooting, this is explicitly noted with a reference to Section 13.

## 1.1 Before You Begin

- Have the unit's serial number ready (printed on the underside label and visible in the ArgoLink app under Settings > Device Info).
- Confirm the firmware version currently installed (Settings > About in the ArgoLink app).
- If possible, have physical access to the unit, as several procedures require observing the status LED or pressing the physical reset button.

## 2. Status LED Reference

The Pro X9 has a single multi-color status LED on the front face. The table below describes each LED pattern and its meaning.

| LED Pattern | Meaning |
|---|---|
| Solid Green | Normal operation; all systems healthy. |
| Blinking Amber (slow, 1 Hz) | Booting up; normal during the first ~45 seconds after power-on. |
| Blinking Amber (fast, 4 Hz) | Firmware update in progress; do not power off. |
| Solid Amber | Degraded operation; one or more non-critical faults present (see Section 3). |
| Blinking Red (slow, 1 Hz) | Connectivity lost; operating in local buffering mode. |
| Blinking Red (fast, 4 Hz) | Critical fault; unit is not processing I/O. See Section 3. |
| Solid Red | Hardware fault detected; contact Support (Section 14). |
| Off (no light) | No power reaching the unit; see Section 5. |
| Blinking Blue | Bluetooth pairing mode active (commissioning). |

## 3. Error Code Reference

Error codes are visible in the ArgoLink app under Device Health > Fault Log, and are also exposed via the local web console and the Webhook Forwarder for integration into external monitoring systems.

| Code | Description | Severity |
|---|---|---|
| E-101 | WAN connectivity lost (Ethernet/Wi-Fi/Cellular all unreachable) | Warning |
| E-104 | DHCP lease acquisition failed on WAN port | Warning |
| E-108 | MQTT broker connection refused (authentication failure) | Warning |
| E-112 | NTP time sync failed; system clock may drift | Info |
| E-205 | Analog input channel out-of-range reading detected | Warning |
| E-210 | Digital input channel flapping (rapid state changes) | Warning |
| E-214 | Relay output failed to switch (no current detected) | Critical |
| E-220 | Modbus RTU device not responding on RS-485 bus | Warning |
| E-301 | Onboard storage nearing capacity (above 90% full) | Warning |
| E-305 | Onboard storage write failure | Critical |
| E-310 | Data buffer sync failed after connectivity restored | Warning |
| E-401 | Firmware update download failed (checksum mismatch) | Warning |
| E-405 | Firmware update failed to boot; automatic rollback triggered | Critical |
| E-501 | Internal temperature exceeds safe operating threshold | Critical |
| E-510 | Power input voltage out of rated range | Critical |
| E-520 | Supercapacitor backup not holding charge (replace unit) | Critical |

# 4. Connectivity Troubleshooting

## 4.1 Symptom: Unit Shows Blinking Red LED (Connectivity Lost)

Likely causes, in order of probability: (1) WAN cable disconnected or faulty, (2) upstream network outage, (3) incorrect network configuration, (4) DNS resolution failure.

- Step 1 — Verify the WAN Ethernet cable is firmly seated at both ends, or confirm the configured Wi-Fi network is broadcasting and within range.
- Step 2 — Check the local web console (https://192.168.1.1 or configured static IP) by connecting directly to the LAN port with a laptop; if reachable, the unit's local system is healthy and the fault is upstream.
- Step 3 — Confirm the site's internet connection is functioning by testing another device on the same network.
- Step 4 — If using DHCP, confirm the DHCP server has available leases; Error E-104 indicates a DHCP failure specifically.
- Step 5 — If using a static IP, verify the gateway and DNS settings under Settings > Network in ArgoLink match the site's network requirements.
- Step 6 — If all of the above check out and connectivity is still lost, proceed to Section 12 to collect diagnostic logs and escalate per Section 13.

## 4.2 Symptom: Error E-108 (MQTT Authentication Failure)

- Step 1 — Confirm the MQTT broker credentials configured in Settings > Integrations > MQTT match the current credentials on the broker side.
- Step 2 — If using certificate-based authentication, confirm the certificate has not expired (visible under Settings > Security > Certificates).
- Step 3 — Confirm the broker's firewall allows inbound connections from the unit's current IP address, especially after a network change.
- Step 4 — Re-issue a new certificate or credential set if the above do not resolve the issue, and apply it through the ArgoLink app.

# 5. Power and Boot Troubleshooting

## 5.1 Symptom: LED Is Completely Off

- Step 1 — Verify the power supply is plugged into a working outlet and the output cable is firmly connected to the unit's power terminal.
- Step 2 — If using PoE+, confirm the upstream switch port is configured for PoE+ (802.3at) and not standard PoE (802.3af), which provides insufficient wattage.
- Step 3 — Measure the voltage at the power terminal with a multimeter; it should read within the 12–24V DC rated range (Product Guide, Section 7).
- Step 4 — If voltage is present but the LED remains off, the unit's internal power supply may have failed; proceed to Section 13 for RMA evaluation.

## 5.2 Symptom: Unit Boot-Loops (LED Cycles Amber Repeatedly)

This typically indicates a firmware corruption or failed update that could not complete a rollback automatically.

- Step 1 — Allow the unit up to 5 minutes; ArgoOS attempts up to 3 automatic rollback cycles before halting.
- Step 2 — If boot-looping continues beyond 5 minutes, connect via the RS-232 console port at 115200 baud to view boot logs.
- Step 3 — If boot logs show a firmware checksum failure, perform the Factory Reset Procedure in Section 8, which reflashes the unit to its last known-good firmware image stored in the recovery partition.
- Step 4 — If the Factory Reset does not resolve the boot-loop, the unit requires RMA; proceed to Section 16.

# 6. Sensor and I/O Troubleshooting

## 6.1 Symptom: Error E-205 (Analog Input Out-of-Range)

- Step 1 — Confirm the connected sensor's output range (e.g., 4-20mA or 0-10V) matches the channel's configured input type in ArgoLink under Settings > I/O Configuration.
- Step 2 — Check wiring polarity at the terminal block; reversed polarity commonly produces out-of-range or zero readings.
- Step 3 — Measure the sensor's actual output directly with a multimeter to isolate whether the fault is in the sensor or the controller.
- Step 4 — If the sensor output is correct but the controller still reports out-of-range, the analog input channel may require recalibration via Settings > I/O Configuration > Calibrate, or the channel's hardware may be faulty.

## 6.2 Symptom: Error E-220 (Modbus RTU Device Not Responding)

- Step 1 — Confirm the RS-485 bus wiring (A/B lines) is not reversed and that termination resistors are correctly placed at both ends of the bus.
- Step 2 — Verify the Modbus device's configured slave address matches the address set in ArgoLink's Modbus device list.
- Step 3 — Confirm baud rate, parity, and stop-bit settings match between the Pro X9's RS-485 port configuration and the connected device.
- Step 4 — Use the built-in Modbus Scanner tool (Settings > Diagnostics > Modbus Scan) to detect any responding devices on the bus and compare against expected addresses.

## 6.3 Symptom: Error E-214 (Relay Output Failed to Switch)

This is a Critical-severity fault. Disconnect power before inspecting wiring.

- Step 1 — Verify the connected load does not exceed the relay's rated switching capacity of 5A at 250V AC (Product Guide, Section 15.1); an overloaded relay may have failed open.
- Step 2 — Inspect the relay terminal for signs of arcing or burn marks, which indicate the relay contact has failed and the unit requires RMA.
- Step 3 — If no visible damage, attempt to manually trigger the relay via Settings > I/O Configuration > Test Relay in ArgoLink and listen for an audible click; absence of a click confirms relay failure.

# 7. Firmware Update Troubleshooting

## 7.1 Symptom: Error E-401 (Firmware Download Checksum Mismatch)

- Step 1 — Retry the update from Settings > Firmware > Check for Updates; transient network issues during download are the most common cause.
- Step 2 — If the error persists across 3 attempts, verify the unit's system clock is accurate (see E-112); an incorrect clock can cause certificate validation failures during the secure download.
- Step 3 — If the issue continues, manually download the firmware image from the Veltrion Support Portal and side-load it via the USB-C recovery port, following the manual update procedure in the Product Guide Section 11.

## 7.2 Symptom: Error E-405 (Firmware Update Failed to Boot, Rollback Triggered)

ArgoOS automatically reverts to the previous working firmware version after 3 failed boot attempts on a new version. No data loss occurs during this rollback, as configuration and buffered sensor data are stored on a separate partition from the firmware image.

- Step 1 — Confirm the rollback completed successfully; the LED should return to solid green and the firmware version shown in ArgoLink should match the prior version.
- Step 2 — Before attempting the update again, verify the unit meets minimum free storage requirements (at least 500 MB free) under Settings > Storage.
- Step 3 — If the same update fails a second time after rollback, do not retry further; escalate to Veltrion Support (Section 13) with the diagnostic logs (Section 12), as this may indicate a hardware-specific incompatibility.

# 8. Factory Reset Procedure

A factory reset restores the unit to its default configuration and reflashes firmware from the onboard recovery partition. This should be used only after other troubleshooting steps have been exhausted, as it clears all local configuration (network settings, rule definitions, sensor mappings) though buffered sensor data history is preserved.

## 8.1 Soft Reset (via ArgoLink App)

- Step 1 — Navigate to Settings > Advanced > Factory Reset.
- Step 2 — Choose "Reset Configuration Only" to preserve firmware version, or "Full Reset" to also reflash firmware from the recovery partition.
- Step 3 — Confirm the action; the unit will reboot and the LED will cycle through the boot sequence described in Section 2.

## 8.2 Hard Reset (Physical Button, for Unresponsive Units)

- Step 1 — Locate the recessed reset button on the unit's rear panel, accessible with a paperclip or similar tool.
- Step 2 — With the unit powered on, press and hold the reset button for 10 seconds until the LED turns solid amber.
- Step 3 — Release the button; the unit will perform a Full Reset and reboot automatically, taking approximately 3 minutes to complete.
- Step 4 — Re-commission the unit following the installation steps in the Product Guide, Section 9.

# 9. Data Buffer and Sync Issues

## 9.1 Symptom: Error E-301 (Storage Nearing Capacity)

- Step 1 — Check connectivity status; a full buffer typically indicates an extended period of lost connectivity (see Section 4).
- Step 2 — Once connectivity is restored, confirm the sync process has started under Settings > Data Buffer > Sync Status; sync rate is approximately 10,000 records per minute under normal conditions.
- Step 3 — If storage remains near capacity after connectivity is restored and 30 minutes have passed, check Error E-310 (sync failure) and proceed to 9.2.

## 9.2 Symptom: Error E-310 (Buffer Sync Failed)

- Step 1 — Verify the configured MQTT broker or REST endpoint is reachable and accepting connections (see Section 4.2 for MQTT-specific checks).
- Step 2 — Check Settings > Data Buffer > Sync Log for the specific rejection reason returned by the receiving endpoint.
- Step 3 — If the endpoint is rejecting data due to schema mismatch, confirm no recent changes were made to the expected data format on the receiving platform.
- Step 4 — As a last resort, buffered data can be manually exported via the local web console (Settings > Data Buffer > Export to CSV) for manual upload.

# 10. ArgoLink App Troubleshooting

## 10.1 Symptom: App Cannot Find Device During Bluetooth Pairing

- Step 1 — Confirm the unit's LED is blinking blue, indicating Bluetooth pairing mode is active (triggered automatically for 10 minutes after power-on, or manually via the reset button held for 2 seconds).
- Step 2 — Confirm the commissioning device's Bluetooth is enabled and location services are granted to the ArgoLink app (required by mobile OS for BLE scanning).
- Step 3 — Move the commissioning device within 3 meters of the unit; BLE pairing range is limited compared to the unit's operational Wi-Fi/Ethernet range.
- Step 4 — Restart the ArgoLink app and retry the "Add Device" flow.

## 10.2 Symptom: App Shows Device as Offline Despite Solid Green LED

- Step 1 — Confirm the mobile device and the Pro X9 unit are communicating through the cloud relay (not directly); check the commissioning device's own internet connectivity.
- Step 2 — Force-close and reopen the ArgoLink app to refresh the device status cache.
- Step 3 — Confirm the unit has not been removed or reassigned to a different account via the Fleet View dashboard (Product Guide, Section 12.2).

# 11. Common Issues and Fixes (Quick Reference Table)

| Symptom | Most Likely Cause | Section |
|---|---|---|
| LED off entirely | No power reaching unit | 5.1 |
| Blinking red LED | Connectivity lost | 4.1 |
| Boot-loop (amber cycling) | Corrupted firmware | 5.2 |
| E-205 analog out-of-range | Wrong channel config or wiring | 6.1 |
| E-214 relay failure | Overloaded or failed relay contact | 6.3 |
| E-220 Modbus device silent | Wiring, address, or baud mismatch | 6.2 |
| E-401 firmware checksum fail | Network interruption during download | 7.1 |
| E-301 storage nearing full | Prolonged connectivity outage | 9.1 |
| App can't find device | Not in pairing mode / BLE range | 10.1 |

# 12. Diagnostic Log Collection

Before escalating any unresolved issue to Veltrion Support, collect a diagnostic log bundle to accelerate root-cause analysis.

## 12.1 Collecting Logs via ArgoLink

- Step 1 — Navigate to Settings > Advanced > Diagnostic Logs.
- Step 2 — Tap "Generate Support Bundle"; this compiles system logs, fault history, and configuration snapshot into a single encrypted .vlog file.
- Step 3 — Share the file directly to a support ticket via the app, or export it for manual email attachment.

## 12.2 Collecting Logs via Local Web Console

For units without app connectivity, logs can be downloaded directly from the local web console at Settings > Diagnostics > Download Logs when connected via the LAN port.

## 13. Escalation Procedure

If a field-level troubleshooting procedure does not resolve the issue, escalate following the tiered structure below.

| Tier | Scope | Target Response Time |
|---|---|---|
| Tier 1 — Site Technician | Procedures in Sections 4–10 of this guide | Immediate (self-service) |
| Tier 2 — Veltrion Remote Support | Log analysis, remote configuration review, RMA initiation | 4 business hours |
| Tier 3 — Veltrion Engineering | Hardware-level or firmware-level defect investigation | 2 business days |

To escalate to Tier 2, open a support ticket via the Veltrion Support Portal with the diagnostic bundle from Section 12 attached, the unit's serial number, and a description of troubleshooting steps already attempted.

## 14. Support Tiers and Contact Channels

| Channel | Availability | Best For |
|---|---|---|
| Support Portal (ticket-based) | 24/7 submission; Tier 2 responds within SLA | Non-urgent issues, RMA requests |
| Phone Support | Mon–Fri, 8:00–20:00 local time | Critical-severity faults (E-5xx codes, Section 3) |
| Live Chat (in ArgoLink app) | Mon–Fri, 9:00–18:00 local time | Quick configuration questions |
| Email | support-mailbox monitored during business hours | Documentation requests, non-urgent follow-ups |

# 15. Service Level Agreements (SLA)

| Severity | Definition | Initial Response | Target Resolution |
|---|---|---|---|
| Critical | Unit non-operational; production impact (E-5xx, E-214, E-305) | 1 hour (phone) | 4 hours (ticket) | 8 business hours |
| Warning | Degraded but operational (most E-1xx, E-2xx, E-3xx codes) | 4 business hours | 2 business days |
| Info | No operational impact (E-112 and similar) | 1 business day | 5 business days |

SLA targets above apply to customers with an active Standard or Premium Support contract. Customers without an active support contract are supported on a best-effort basis via the Support Portal only.

# 16. Return Merchandise Authorization (RMA) Process

## 16.1 When to Request an RMA

- Hardware fault confirmed per Sections 5–6 that cannot be resolved via factory reset or firmware recovery.
- Visible physical damage not caused by improper installation or misuse.
- Repeated Critical-severity faults (E-5xx) after all field troubleshooting steps have been exhausted.

## 16.2 RMA Steps

- Step 1 — Open a Tier 2 support ticket (Section 13) and request RMA evaluation, including diagnostic logs and a description of the fault.
- Step 2 — Veltrion Support reviews the logs and, if RMA is approved, issues an RMA number and prepaid return shipping label (for units within warranty, see Product Guide Section 19).
- Step 3 — Package the unit in its original packaging where possible, or equivalent anti-static, shock-protective packaging.
- Step 4 — Ship the unit with the RMA number clearly marked on the exterior label.
- Step 5 — Veltrion inspects the returned unit within 5 business days of receipt and issues a repaired or replacement unit, or a denial notice with explanation if the fault is found to be outside warranty scope.

## 17. Preventive Maintenance Checklist

Veltrion recommends the following maintenance checks on a quarterly basis to reduce the likelihood of field faults.

- Inspect the enclosure gasket and connector caps for wear, confirming the IP65 seal remains intact.
- Verify all terminal block wiring connections remain firmly seated (vibration over time can loosen terminals).
- Review the Fault Log in ArgoLink for any recurring Warning-severity codes that may indicate a developing issue.
- Confirm the unit is running the latest Stable-channel firmware (Section 7 in the Product Guide).
- Check onboard storage utilization under Settings > Storage; proactively investigate if consistently above 70%.
- Test relay outputs via the Settings > I/O Configuration > Test Relay function to confirm switching behavior remains audible and responsive.

# 18. Frequently Asked Questions

## Q1. My unit shows a Warning-severity error but seems to be working fine — should I still act on it?

Warning-severity codes do not indicate an immediate operational failure, but left unaddressed they can develop into Critical faults. Follow the relevant section's resolution steps during your next scheduled maintenance window.

## Q2. How do I know if my issue needs a Factory Reset or an RMA?

Attempt a Factory Reset (Section 8) first for any software- or configuration-related symptom. If the same fault persists immediately after a Full Factory Reset, this strongly suggests a hardware fault and an RMA request (Section 16) is appropriate.

## Q3. Will a Factory Reset erase my historical sensor data?

No. Configuration settings (network, rules, sensor mappings) are cleared, but buffered and already-synced sensor data history is stored on a separate partition and is preserved through both Soft and Hard resets.

## Q4. Can I troubleshoot a unit remotely without physical access?

Most Warning-severity issues (Sections 4, 7, 9, 10) can be diagnosed and resolved remotely via the ArgoLink app's Fleet View. Sections requiring physical inspection (Sections 5, 6, 8.2) require someone on-site.

## 19. Glossary

| Term | Meaning |
|---|---|
| Fault Log | The in-app record of error codes raised by the unit, with timestamps. |
| Recovery Partition | A protected storage area holding the last known-good firmware image. |
| RMA | Return Merchandise Authorization; the formal process for returning a faulty unit. |
| Support Bundle | An encrypted .vlog file containing logs and configuration for diagnostics. |
| Tier 1/2/3 | Veltrion's escalation levels for support, from site-level to engineering. |

*All product names, error codes, and company details in this document are fictional and created for software testing and training purposes.*